Topic 3: Operators and Expressions

Task 1: Restaurant bill calculator

```
let itemPrice = 450.0
let quantity = 3
let subtotal = itemPrice * Double(quantity)
let hasCoupon = true
let discount = (hasCoupon && subtotal > 1000) ? subtotal * 0.10 : 0.0
let discountedAmount = subtotal - discount
let gst = discountedAmount * 0.18
let total = discountedAmount + gst
let people = 4
let share = total / Double(people)
print("Subtotal: \(subtotal)")
print("Discount: \(discount)")
print("GST: \(gst)")
print("Total: \(total)")
print("Each person pays: \(share)")
let amount = 1234
let hundreds = amount / 100
let remainder = amount % 100
print("\(hundreds) hundreds and \(remainder) rupees left")
```

```
Subtotal: 1350.0
Discount: 135.0
GST: 218.7
Total: 1433.7
Each person pays: 358.425
12 hundreds and 34 rupees left
```

Task 2: Leap year and range checks

```
let years = [1900, 2000, 2024, 2026]
for year in years {
    let isLeap = (year % 4 == 0 && year % 100 != 0) || year % 400 == 0
    print("\(year) leap year? \(isLeap)")
}
let number = 57
let inRange = (1...100).contains(number)
print("\(number) between 1 and 100? \(inRange)")
let n = 24
let evenNotTen = n % 2 == 0 && n % 10 != 0
print("\(n) even and not multiple of 10? \(evenNotTen)")
let nickname: String? = nil
print(nickname ?? "Guest")
```

```
1900 leap year? false
2000 leap year? true
2024 leap year? true
2026 leap year? false
57 between 1 and 100? true
24 even and not multiple of 10? true
Guest
```

Task 3: Predict the output

```
print(7 / 2)
print(7 % 2)
print(-7 % 3)
print(7.0 / 2)
print(10 - 2 * 3 + 4 / 2)
print(5 > 3 && 2 > 4 || true)
```

```
print(1 << 3)
print(6 & 3, 6 | 3, 6 ^ 3)
let x: UInt8 = 255
print(x &+ 1)
```

```
3
1
-1
3.5
6
true
8
2 7 5
0
```